

have a voltage difference, and are then connected by a wire, current flows so as to equalise that voltage difference.

Sounds like a recipe for a battery. Why not just use capacitors? The problem is they can't hold much charge. They'd have to be *huge* to hold as much charge as a standard battery. Why? Because the surface area of the conductive surfaces (the electrodes) would have to be very large.

Enter the nanotube, to the sound of a boisterous orchestra. Joel Schindall, an engineer at MIT, thought up this one: cover the electrodes with millions of nanotubes. And how would this work, please?

We said the surface area of the electrodes would have to be much larger. Now, why does a Turkish towel—with its coating of water-absorbing threads—work better than a plain cloth towel? Because it has more surface area. Just so, the nanotubes increase the surface area of the electrodes to a very high degree, without taking up too much space themselves. The result is a capacitor that can rival a traditional battery in terms of charge.

Timeline: five years. (Schindall's estimate, not ours.)

Motorola And Their "Hybrid" Display

We begin with a disclaimer here: the use of nanotubes in displays does not promise world peace. It doesn't even promise to revolutionise displays—not too many companies have yet bought into the idea. But we must mention this one so you get an idea of the sheer range of applications nanotubes have!

There are too many new display technologies, and nanotubes have jumped in to add to the confusion. Will they, won't they deliver? Motorola's nano-emissive display design has been touted (by Motorola—who else?) to be superior to all flat-panel technologies. In what way? Well, all the usual suspects: very good brightness, very quick response times, colours almost as good as those on a CRT, longer life, and—the best part—they're cheaper to manufacture.

How does it work? If you remember engineering college lessons—or even if you don't—it's simple. CRTs sweep an electron beam across the screen, which is coated with phosphors that glow when excited by electrons. The nano-emissive display uses an array of nanotubes to fire the electrons, instead of whatever fires them in regular CRTs, and there are clusters of nanotubes behind each pixel. Think of an NED as a cross between CRT and flat-panel technologies.

Motorola's prototype is 4.7 inches diagonally, with a resolution of 128 x 96 pixels. It was designed to be one piece of a 42-inch HDTV screen with a resolution of 1280 x 720. Well, they're getting there.

There's always a catch to any nanotube application, and it always goes like this: "how to get the nanotubes to work in <some> way." Here, the problem is attaching the nanotubes to a glass substrate. Motorola says they're managing pretty well.

Nantero...

...is a small company that has big plans for NRAM (nano RAM), saying it will replace all

There's always a catch to any nanotube application, and it always goes like this: "how to get the nanotubes to work in <some> way"

kinds of memory altogether, including the hard disk, once and for all. They say we'll have a universal memory of sorts in the form of NRAM. It's again about nanotech coming to the rescue and making possible the dream of Universal Memory.

There happen to be several contenders for Universal Memory—notably MRAM ("M" for "magnetic"), FRAM ("F" for "Ferroelectric Materials"), and more. We're hoping to see something concrete soon—these terms have been bandied about for too long now! In any case, to understand how NRAM works doesn't require a PhD.

Refer the figure below, and look at each component, specially the interconnects, the ribbon-like things, and the grey electrodes. The nanotube ribbons, anchored to the interconnects, stay suspended over the electrodes. Now when a voltage is applied between the ribbon and the electrode, the ribbon *bends and touches the electrode*.

That's one key to the working. The other key is that the ribbon stays stuck to the electrode even when the voltage is turned off, due to certain molecular surface attraction forces. This makes for non-volatility, meaning that power need not be continuously supplied to the device.

And thus are obtained the ones and zeroes: when the ribbon is far from the electrode, it's a zero. When the ribbon is touching the electrode, the resistance between them is much lower, and it's a one.

How does this compare with SRAM and DRAM? It turns out that the switching can be as fast as that of SRAM, making NRAM have the "fast" characteristic of SRAM; at the same time, the density of the assembly approaches that of DRAM. And the whole thing is non-volatile, like we mentioned, so it's got the goodness of Flash.

NRAM, in sum, trounces all three—DRAM, SRAM, and Flash—except for one little thing: it isn't as fast as SRAM. So we have the best of all three worlds, give and take a little. Visit nantero.com for more on NRAM. There's a movie illustrating the principles of operation of NRAM.

